A PROJECT REPORT

ON

Dynamic Thermal Modelling and Energy-Efficient Cooling Optimization of AI Data Centres Using Optimal Control

BY

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Date of submission:

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Discipline of the student: Mathematics and Computing

Course Instructor: Amit Setia

Keywords:

Project area(s):

Acknowledgments:

Signature of the student
Date

Signature of the faculty member
Date

Abstract

Write a concise summary of the project. The abstract should briefly state:

- the mathematical method being studied,
- the application considered,
- the main approach or methodology,
- and the principal outcome or observation.

Chapter 1

Mathematical Methods: Your Topic

1.1 Literature Review

Briefly discuss the mathematical method and relevant references. Include the books, papers, lecture material, or other sources that have been consulted, and cite them appropriately.

1.2 Mathematical Method in Your Topic

Describe the mathematical method used in the project. Include, as appropriate:

- important definitions,
- mathematical formulation,
- relevant formulas,
- algorithms or computational steps,
- important properties or theorems,
- and a brief justification for using the method.

1.3 Examples of the Mathematical Method

Give **at least two mathematical examples** demonstrating the method. The examples should be worked out clearly and should help the reader understand the mathematical technique before its application is introduced.

Chapter 2

An Application of the Mathematical Method in Your Topic

2.1 Literature Review: Introduction to the Problem/Application

Introduce the real-world problem or application. Explain:

- what the problem is,
- where it arises,
- and why it is important.

2.2 Where and Why the Method is Used

Explain where the mathematical method enters the application and why it is appropriate for the problem under consideration.

2.3 Methodology: How is the Method Used in the Application?

Explain how the mathematical method is applied to the problem. Include the mathematical formulation or model and clearly connect the theory from Chapter 1 to the application.

Where appropriate, include:

- equations or matrix formulations,
- algorithms,
- numerical procedures,
- assumptions,
- datasets or sample inputs,
- and computational steps.

2.4 Numerical/Computational Demonstration and Results

Present the output of the project. This may include a worked example, numerical computation, simulation, graph, image, table, or other appropriate demonstration.

The output should clearly show how the mathematical method is being used.

Chapter 3

Conclusions and Future Work

Interpret the results obtained in the project and summarize the main conclusions.

The discussion should include:

- what was learned from the mathematical analysis,
- what the numerical or computational results show,
- the strengths or limitations of the chosen method,
- and possible improvements or extensions.

Include one or two paragraphs on possible future work.

Generative AI Usage Declaration

Disclose whether any Generative AI tool was used while preparing the project report.

If Generative AI tools were used, provide the following information:

- Name of the Generative AI tool(s) used.
- Purpose for which the tool(s) were used (for example, language refinement, coding assistance, or clarification of concepts).
- A brief description of the extent of such use.

Note: AI-generated images, figures, diagrams, or illustrations are strictly prohibited.

If no Generative AI tool was used, write:

"I declare that no Generative AI tool was used in preparing this project report."

A brief statement can be prepared as follows:

Statement: During the preparation of this report, I have used [NAME OF TOOL / SERVICE] in order to [REASON]. After using this tool/service, I reviewed and edited the content as needed and take full responsibility for the content of the submitted report.

Bibliography

List all books, papers, websites, lecture notes, datasets, software/documentation, or other sources referred to in the report. Use a consistent citation style.

- Author Name, *Title of Book/Paper*, Publisher/Journal, Year.
- Author Name, "Title of Article," *Journal Name*, Volume, Issue, Pages, Year.

Appendix A

Programming Code

Include all relevant programming code used in the project. Add brief comments explaining the purpose of each code segment.